Credit Card Fraud & Transaction Risk Analysis

Fraud frequency, financial impact and risk-monitoring dashboard

Tools: Python ● Pandas ● Seaborn ● Power BI ● DAX

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1. Company Background

This project is based on an **illustrative business scenario**: a retail card issuer ("the Bank") that provides credit cards to customers and processes thousands of card transactions every hour. Every transaction is either genuine or, rarely, fraudulent.

The Bank earns income from card usage but loses money and customer trust when fraudulent transactions are approved. The data used here is a public, anonymised dataset of European cardholders (published by the ULB Machine Learning Group), so no real company name or private bank data is used.

2. Business Problem

Fraud is **rare but costly**. In this dataset only about 1 in 600 transactions is fraudulent, so fraud is almost invisible in normal volume reports. At the same time, the Bank cannot manually review every transaction, and blocking genuine customers damages customer experience.

The fraud and risk team therefore needs clear answers to three questions:

- **How often** does fraud happen? (frequency)
- **How much money** is involved? (financial impact)
- **Where** is the risk concentrated? (amount sizes and time windows)

This project turns raw transaction data into fraud KPIs, an interactive Power BI monitoring dashboard and practical recommendations. It is a **descriptive risk-analytics project**, not a machine-learning prediction model.

3. Business Objectives

#	Objective	Measured by	Result
1	Measure fraud frequency and financial impact	Fraud Rate %, Fraud Value Rate %	0.167% of transactions; 0.233% of value
2	Compare fraud and legitimate amount behaviour	Average, median, 90th percentile	Fraud median 9.82 vs 22.00; fraud average 123.87 vs 88.41
3	Find where fraud cases and fraud money are concentrated	% of fraud cases and % of fraud value by amount band	"0 to 10": 50.3% of cases, 0.8% of value; "500+": 7.2% of cases, 52.4% of value
4	Check whether fraud rate changes over the data window	Fraud rate by 6-hour elapsed-time period	Ranges from 0.076% to 0.520%
5	Identify anonymised features statistically linked to fraud	Correlation with Class	V17, V14, V12 strongest
6	Define monitoring KPIs and give recommendations	KPI dashboard and recommendation list	5 headline KPI cards on the dashboard (10 KPIs calculated in Python), 7 recommendations

4. Stakeholders

Stakeholder	What they need to decide	Dashboard view used
Fraud Operations Manager	Which alerts to review first; reviewer workload	Fraud by amount band, time period, Top 10 fraud table
Head of Risk	Loss exposure and strength of controls	Fraud value, fraud value rate, concentration chart

Stakeholder	What they need to decide	Dashboard view used
Card Product Manager	How much customer friction is acceptable	Fraud rate by amount band
Senior Management	Overall fraud health of the portfolio	KPI cards
Risk Analytics / Data Science team	Where to build detection rules	Feature analysis, deep-dive page
Customers	Safe card use with few false blocks	Indirect (benefit from better controls)

5. Business KPIs

KPI	Formula	Value
Total Transactions	Count of all transactions	283,726
Fraud Transactions	Count where Class = 1	473
Legitimate Transactions	Count where Class = 0	283,253
Fraud Rate %	Fraud Transactions ÷ Total Transactions × 100	0.167%
Total Transaction Value	Sum of Amount	25,102,001.68
Fraud Transaction Value	Sum of Amount where Class = 1	58,591.39
Fraud Value Rate %	Fraud Value ÷ Total Value × 100	0.233%
Average Transaction Amount	Total Value ÷ Total Transactions	88.47
Average Fraud Amount	Fraud Value ÷ Fraud Transactions	123.87
Maximum Fraud Amount	Maximum Amount where Class = 1	2,125.87

All 10 KPIs are calculated in Python. The Power BI dashboard shows the 5 headline KPIs as cards (Total Transactions, Fraud Transactions, Fraud Rate %, Fraud Transaction Value, Fraud Value Rate %).

Four terms to keep separate: transaction count (how many) · fraud rate (how often) · fraud amount (how much money) · fraud value rate (fraud money as a share of all money).
Fraud value rate (0.233%) is higher than fraud rate (0.167%) because fraud transactions are larger on average.

6. Domain Knowledge

Key banking and fraud terms needed to understand this project:

Term	Meaning
Issuer	The bank that issues the card to the customer and bears fraud losses (our "Bank").
Authorisation	The real-time decision to approve or decline a card transaction, taken in a fraction of a second.
Card-not-present (CNP) fraud	Fraud using stolen card details online or by phone, without the physical card.
Card testing	Fraudsters try very small purchases to check a stolen card works before larger fraud. This is a possible explanation for many small-value frauds, but cannot be confirmed from this dataset.
Chargeback	A customer dispute that returns money for a transaction they say they did not make.
False positive	A genuine transaction wrongly flagged or blocked. It creates customer friction.
Velocity check	A rule that flags many transactions on one card in a short time.

Term	Meaning
Step-up authentication	Asking for extra verification (for example a one-time code) only on riskier transactions.
Manual review queue	Alerts that a human analyst checks. Limited by staff capacity.
Class imbalance	One class (fraud) is far smaller than the other. Accuracy becomes a misleading metric.
PCI DSS	Global security standard for organisations that handle card data.

Common industry challenges: fraud is rare, fraud patterns change, customers expect fast approvals, and review teams have limited capacity. Regulations and customer-liability rules differ by country and change over time, so they should be checked for the relevant market before real-world use.

7. Dataset Overview

Item	Detail
Source	Credit Card Fraud Detection dataset, ULB Machine Learning Group (Kaggle). European cardholders, September 2013.
Records / features	284,807 transactions · 31 columns
Time covered	About 48 hours (172,792 seconds)
Data types	All 31 columns numeric
Missing values	0
Exact duplicate rows	1,081 (19 fraud) → removed. Cleaned data: 283,726 rows, 473 fraud
Target variable	Class: 0 = legitimate, 1 = fraud
Class balance	Fraud = 0.167% (extreme imbalance)

8. Data Dictionary

Column	Description
Time	Seconds elapsed since the first transaction in the dataset. Not clock time.
V1 – V28	Anonymised numerical features produced by PCA (a mathematical transformation applied by the data owners to protect confidentiality). They have no known business meaning; they must not be described as age, merchant, location, etc.
Amount	Transaction amount (currency not stated in the dataset).
Class	Target label. 0 = legitimate, 1 = fraud.
Transaction_Type *(created)*	"Legitimate" or "Fraud", readable version of Class.
Time_Hours *(created)*	Time ÷ 3600: elapsed hours.
Time_Period *(created)*	Eight 6-hour elapsed-time bins: 0 to 6 hrs … 42 to 48 hrs. Not real-world clock time.
Elapsed_Hour *(created)*	Whole elapsed hour, 0 to 47.
Amount_Band *(created)*	0 to 10, 10 to 50, 50 to 100, 100 to 500, 500+ (upper edge included).
*_Order columns *(created)*	Numbers 1, 2, 3… used so Power BI sorts bands and periods correctly.

9. Business Questions

Instead of simple "what is the average?" queries, the analysis answers business questions:

Business question	Answer from the data
Where is fraud money concentrated compared with fraud cases?	Small transactions ("0 to 10") hold 50.3% of fraud cases but 0.8% of fraud value. Transactions above 500 hold 7.2% of cases but 52.4% of value.
Which transaction sizes need the strictest controls?	"500+" has the highest fraud rate (0.373%, 2.2× average). Bands above 100 hold 88.3% of fraud value.
Are fraud transactions typically larger than genuine ones?	Not typically: median fraud is 9.82 vs 22.00. But a few large frauds lift the average (123.87 vs 88.41).
Does fraud risk change across the 48-hour window?	Yes. Rate ranges from 0.076% to 0.520%; the highest-rate periods are also the quietest in volume.
How much of the loss comes from a few cases?	The top 5 frauds are 14.1% of fraud value; 34 frauds above 500 hold 52.4%.
Which anonymised signals separate fraud from genuine?	V17, V14, V12 (negative correlation with fraud). Statistical only.
What should a fraud team monitor every week?	Fraud rate, fraud value rate, average/max fraud amount, fraud concentration by amount band.

10. Data Cleaning & Preparation (Python)

The analysis was done in a Google Colab notebook using Python, Pandas, NumPy, Matplotlib and Seaborn. The raw file was uploaded once and kept unchanged; all cleaning was done on a copy.

Data quality check

Check	Result	Decision and reason
Missing values	0	Nothing to fix
Non-numeric columns / invalid Class / negative Amount or Time	0	Nothing to fix
Exact duplicate rows	1,081 (19 fraud)	Removed. Rows identical in all 31 columns are almost certainly repeated records and would double-count transactions.
Zero-amount transactions	1,825 (27 fraud)	Kept. Zero is a valid value and does not distort totals. Meaning cannot be confirmed.
Class imbalance	0.167% fraud	No resampling. Analytics project, so rates are used instead of accuracy.

Result: 284,807 → **283,726** rows; fraud cases 492 → **473**.

Feature engineering

- **Transaction_Type** from Class (Legitimate / Fraud).
- **Time_Hours, Time_Period, Elapsed_Hour** from Time. The data covers 48 hours, so eight 6-hour bins were needed. These are elapsed-time bins, not clock time.
- **Amount_Band**: 0 to 10, 10 to 50, 50 to 100, 100 to 500, 500+. Bands were chosen after inspecting the amount distribution (median about 22, most values below 100, very high amounts rare).
- Eight CSV files were exported and loaded into Power BI.

11. Exploratory & Statistical Analysis (Python)

11.1 Extreme class imbalance

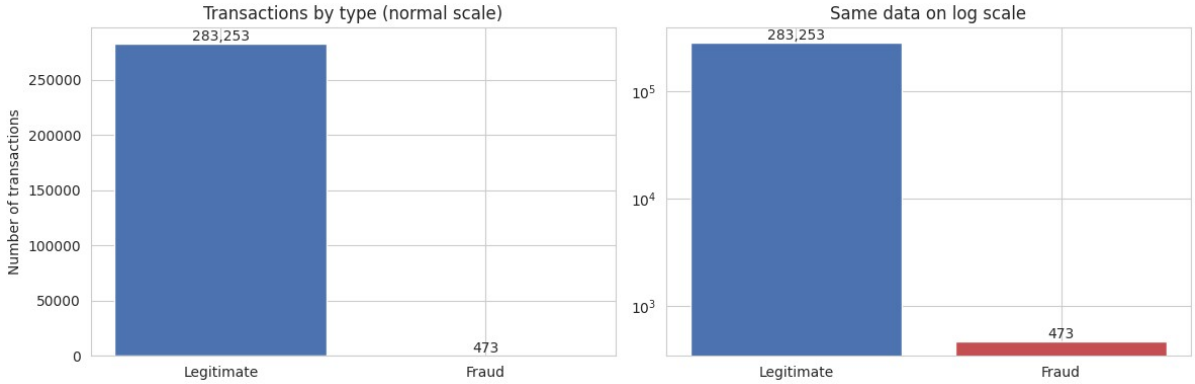


Figure 1: Fraud is a very thin slice of all transactions (normal scale vs log scale)

Only 473 of 283,726 transactions are fraud (0.167%). A report that labels every transaction "legitimate" would be 99.83% "accurate" and catch no fraud, so accuracy is not used. Rates and values are used instead.

11.2 Fraud vs legitimate amounts

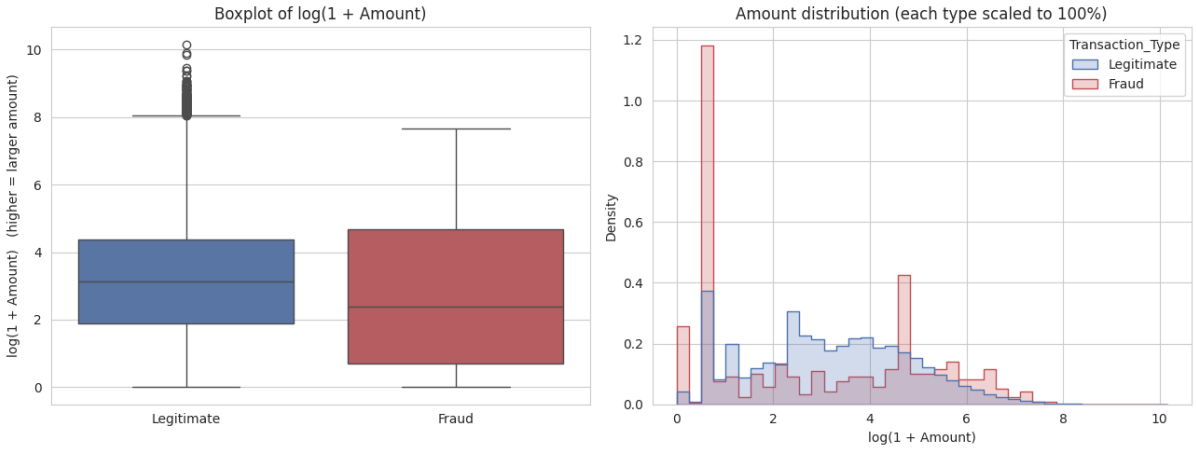


Figure 2: Amount comparison (log scale because amounts are highly skewed)

	Transactions	Average	Median	90th percentile	Maximum
Fraud	473	123.87	9.82	348.65	2,125.87
Legitimate	283,253	88.41	22.00	203.00	25,691.16

Half of the fraud cases are below 9.82, yet the 90th percentile of fraud (348.65) is higher than for legitimate transactions (203.00). Fraud has **more small and more large amounts** than genuine activity.

11.3 Fraud frequency vs financial impact

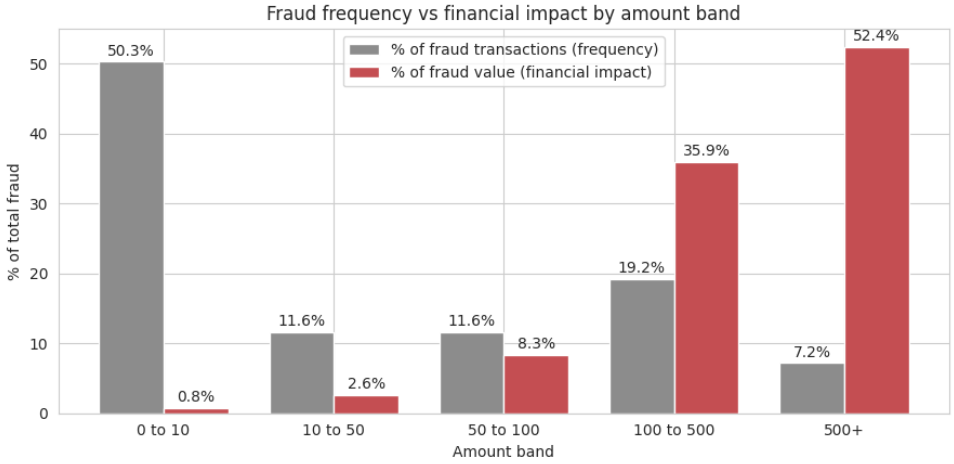


Figure 3: Where fraud cases are vs where fraud money is, by amount band

Amount band	Transactions	Fraud	Fraud rate	% of fraud cases	% of fraud value
0 to 10	99,821	238	0.238%	50.3%	0.8%
10 to 50	90,327	55	0.061%	11.6%	2.6%
50 to 100	37,179	55	0.148%	11.6%	8.3%
100 to 500	47,290	91	0.192%	19.2%	35.9%
500+	9,109	34	0.373%	7.2%	52.4%

This is the key finding. Small transactions create most fraud **cases** (workload), but large transactions hold most of the fraud **money** (loss). The two need different controls.

11.4 Statistical analysis: correlation with Class

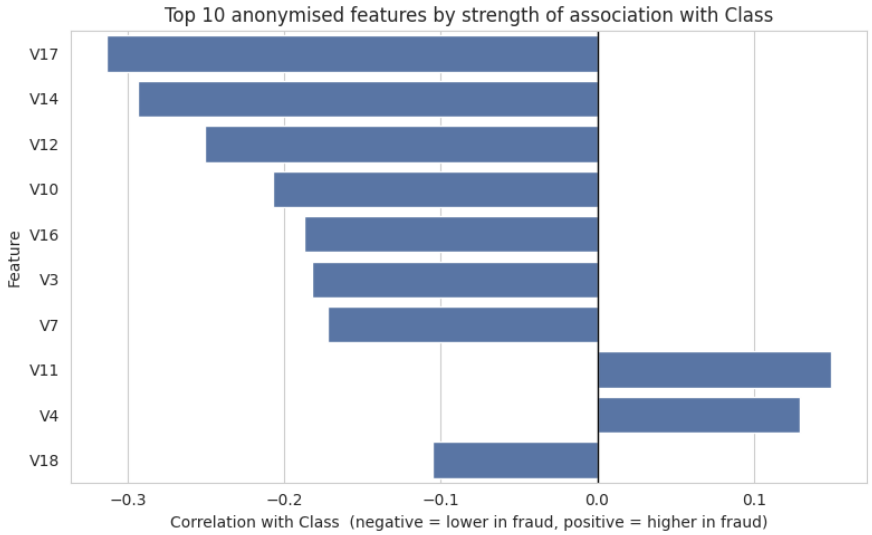


Figure 4: Top 10 anonymised features by strength of association with Class

Feature	Correlation with Class	Reading
V17	-0.313	Lower in fraud
V14	-0.293	Lower in fraud
V12	-0.251	Lower in fraud
V10	-0.207	Lower in fraud
V16	-0.187	Lower in fraud

Correlation shows a **statistical association**, not a cause. V1–V28 are anonymised PCA components, so the analysis identifies statistical patterns rather than directly interpretable business attributes.

12. Power BI Dashboard

The cleaned data was loaded into Power BI. DAX measures were created for fraud rate, fraud value rate and the other KPIs, and the dashboard was validated against the Python results (283,726 transactions, 473 fraud, 58,591.39 fraud value).

- **Page 1 – Executive Overview:** 5 KPI cards (Total Transactions, Fraud Transactions, Fraud Rate %, Fraud Transaction Value, Fraud Value Rate %), Fraud vs Legitimate transactions, Fraud rate by amount band, Fraud frequency vs financial impact, Fraud rate by elapsed time, Amount distribution (average, median, P90), and slicers for Amount Band and Time Period.
- **Page 2 – Fraud Analysis (deep dive):** Fraud rate by elapsed hour, Fraud frequency vs financial impact, Average vs median vs P90 amount, and the Top fraudulent transactions table.
- **Page 3 – Insights & Risk Recommendations:** key findings and recommendations summarised for stakeholders.

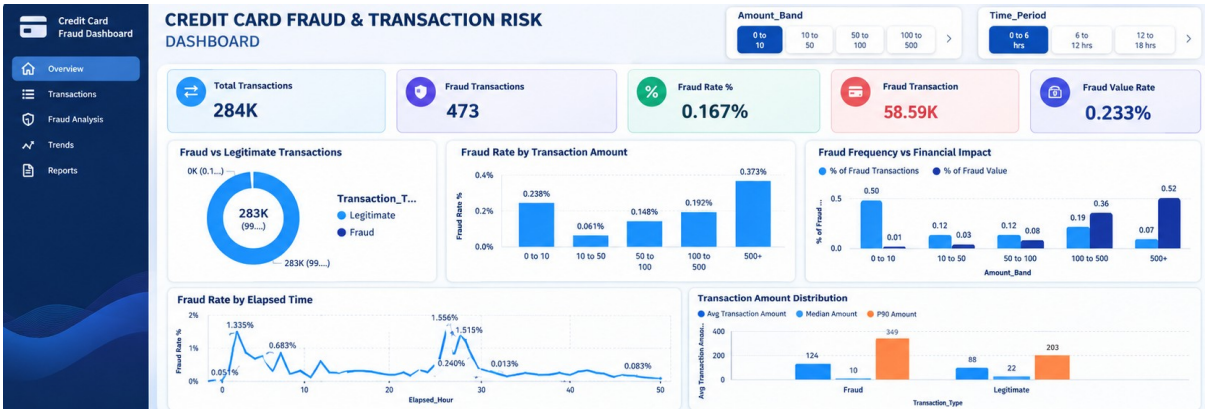


Figure 5: Power BI dashboard, Page 1 (Executive Overview)



Figure 6: Power BI dashboard, Page 2 (Fraud Analysis). Time is elapsed since the first transaction, not clock time.

13. Key Business Insights

#	Insight	Evidence
1	Fraud is rare but real	473 of 283,726 transactions = 0.167% (about 1 in 600).
2	Accuracy is a misleading KPI	"Everything is legitimate" would be 99.83% right and catch 0 frauds.
3	Fraud hurts more by value than by count	Fraud value rate 0.233% vs fraud rate 0.167%.
4	Typical fraud is small; a few large cases lift the average	Median fraud 9.82 vs 22.00 legitimate; average 123.87 vs 88.41.
5	Small transactions: many cases, little money	"0 to 10" = 50.3% of fraud cases, 0.8% of fraud value.
6	Large transactions: highest rate and most money	"500+" = 3.2% of transactions, fraud rate 0.373% (2.2× average), 52.4% of fraud value.
7	Bands above 100 carry most of the loss	88.3% of fraud value from 26.4% of fraud cases.

#	Insight	Evidence
8	Fraud rate varies by elapsed period	0.076% (30 to 36 hrs) to 0.520% (24 to 30 hrs). The highest-rate periods also have the lowest volume (about 12,000 transactions vs 35,000 average).
9	A few large frauds matter	Largest fraud 2,125.87; 34 frauds above 500; top 5 = 14.1% of fraud value.
10	Some anonymised features are statistically linked to fraud	V17 (−0.313), V14 (−0.293), V12 (−0.251).

14. Business Recommendations

These are suggestions based on the observed patterns. They are **not guaranteed** to reduce fraud and should be piloted or back-tested before full rollout.

#	Recommendation	Based on	Owner	Success measure
1	Create a priority alert queue for transactions above 500, using automated checks first and manual review only for the riskiest	Insight 6, 9	Fraud Ops Manager	% of fraud value caught; review time
2	Apply step-up authentication (extra verification) for transactions above 100 instead of blocking	Insight 7	Head of Risk / Product Manager	Fraud value rate; customer drop-off
3	Handle small-value fraud with automated velocity rules, not manual review	Insight 5	Fraud Ops / Analytics	Fraud count in small bands; alert volume
4	Monitor fraud rate by time window with thresholds; confirm with real timestamps before changing staffing	Insight 8	Fraud Ops Manager	Rate vs threshold; time to detect a spike
5	Test anomaly rules on the real attributes behind the strongest statistical signals	Insight 10	Risk Analytics	Share of alerts that are truly fraud
6	Run the KPI dashboard weekly: fraud rate, fraud value rate, average/max fraud amount, concentration	Insights 1–3	Head of Risk	Dashboard refreshed on schedule
7	Capture merchant, location, channel, real timestamp and a unique transaction ID	Limitations, duplicates	Data Engineering	Data completeness; duplicate rate

15. ROI / Business Impact

A real return on investment cannot be calculated without costs, review capacity and customer-friction data. The table below is an **illustrative scenario** using only numbers from this dataset (about 48 hours of transactions, amount units).

The "500+" band has 9,109 transactions (3.2% of all), 34 frauds and **30,682.65 of fraud value (52.4% of all fraud value)**. Suppose a control (such as step-up authentication) stops part of the fraud in this band:

Assumed share of 500+ fraud value prevented	Fraud value protected (48 hours of data)
10%	3,068.27
25%	7,670.66
50%	15,341.33

The trade-off (be honest about it): 9,075 of the 9,109 transactions above 500 (99.6%) are genuine. Only about 1 in 268 is fraud. Manual review of every one of them would be inefficient and extra verification on all of them creates

customer friction.
That is why the recommendation is a **risk-ranked** priority queue and step-up checks, not blanket blocking. Real value = fraud prevented − review cost − lost genuine sales.

Other business benefits: faster detection of fraud-rate spikes, better use of limited reviewer time, one agreed set of fraud KPIs, and clearer data requirements for future analysis.

16. Limitations

- V1–V28 are anonymised PCA features; no business meaning can be given to them.
- Time is elapsed time, so no conclusions about hour-of-day, weekday or night/day are possible.
- Only about 2 days of data and 473 fraud cases after de-duplication. Hourly results are noisy and may not generalise.
- No merchant, customer, location or channel information, so no customer or merchant segmentation.
- Amount currency is not stated, and the dataset is historic (2013) and from one region. The 0.167% rate must not be presented as an industry fraud rate.
- Correlation is association, not causation. This is descriptive analytics, not a detection model.

17. Business Analyst Deliverables

17.1 Business Requirement summary (BRD)

Item	Detail
Business need	A trusted, repeatable view of fraud frequency, fraud value and fraud concentration for risk monitoring.
Scope (in)	Descriptive fraud analytics (Python), KPI definitions, Power BI dashboard, recommendations.
Scope (out)	Fraud prediction models, real-time scoring, customer-level investigation.
Success criteria	KPIs match between Python and Power BI; stakeholders can answer the 7 business questions from the dashboard.
Assumptions	Class labels are correct; duplicates are repeated records; elapsed time is the only time field.

17.2 Functional requirements (FRD summary)

ID	Requirement
FR-1	Dashboard shall show 5 headline KPI cards: total transactions, fraud transactions, fraud rate, fraud transaction value and fraud value rate. The remaining KPIs (total value, average and maximum fraud amount) are calculated in Python.
FR-2	Dashboard shall show fraud rate by amount band and by elapsed-time period.
FR-3	Dashboard shall show % of fraud cases vs % of fraud value by amount band.
FR-4	Dashboard shall list the Top 10 fraudulent transactions by amount.
FR-5	Users shall filter all visuals by Amount Band and Time Period using slicers.
FR-6	All time visuals shall state that time is elapsed, not clock time.

17.3 User stories and acceptance criteria

User story	Acceptance criteria
As a Fraud Operations Manager, I want to see fraud rate by amount band so that I can decide which transactions to review first.	Chart shows 5 bands in correct order; "500+" shows 0.373%; slicers update the chart.
As the Head of Risk, I want fraud value and fraud value rate so that I can understand financial exposure.	Cards show 58,591.39 and 0.233% and match the Python output.
As a Product Manager, I want to see which bands hold most fraud money so that I limit customer friction to where it matters.	Chart shows 52.4% of fraud value in "500+" and 0.8% in "0 to 10".
As a Senior Manager, I want one page of KPIs so that I can check fraud health in under a minute.	Page 1 shows the 5 headline KPI cards with readable formatting and consistent colours.

17.4 As-Is vs To-Be process

	As-Is	To-Be
Monitoring	Fraud is seen only in total volume reports, so it is easy to miss.	Fraud rate and fraud value rate tracked on a dashboard with defined KPIs.
Prioritisation	All transactions treated alike.	Review and verification focused on high-value bands.
Small-value fraud	No specific treatment.	Automated velocity rules.
Data	Duplicates and missing context (merchant, location, timestamp).	De-duplication rules; richer data captured.

To-Be process flow: Transaction data → de-duplication and quality check → fraud KPIs and amount/time bands → dashboard → weekly review → alerts and rule changes → re-measure KPIs.

17.5 Stakeholder matrix

Stakeholder	Influence	Interest	How to engage
Head of Risk	High	High	Weekly KPI review; approves controls
Fraud Operations Manager	Medium	High	Uses dashboard daily; feedback on queues
Card Product Manager	Medium	Medium	Consulted on customer friction
Senior Management	High	Medium	Summary KPIs
Risk Analytics team	Medium	High	Owns data and rule testing

17.6 Risk register

Risk	Impact	Mitigation
Anonymised features cannot be explained	Limited business insight	Use them only as statistical signals; request real attributes
Few fraud cases (473)	Unstable rates in small groups	Report counts with rates; avoid over-reading hourly results
Elapsed time mistaken for clock time	Wrong staffing decisions	Label every time chart; use real timestamps in production
Blocking genuine customers (false positives)	Customer dissatisfaction	Use step-up checks; track false-positive ratio
Duplicates distort KPIs	Overstated counts	De-duplication rule documented and applied

Risk	Impact	Mitigation
Fraud patterns change over time	Dashboard becomes outdated	Refresh data and review thresholds regularly

18. Conclusion, Tools & Deliverables

This project shows how analytics can support fraud-risk decisions even when fraud is very rare. The key lesson is that **fraud frequency and fraud money tell different stories**: small transactions generate most fraud cases, while large transactions hold most of the fraud value. A monitoring dashboard built on fraud rate, fraud value rate and concentration gives the risk team a more useful view than total volumes or accuracy. Recommendations are based on observed patterns and should be tested before rollout.

Tools & technologies

- **Python (Google Colab):** data quality checks, cleaning, feature engineering, EDA, correlation analysis
- **Pandas, NumPy:** data manipulation · **Matplotlib, Seaborn:** visualisation
- **Power BI and DAX:** KPI measures, interactive dashboard, slicers

Project deliverables

- Credit_Card_Fraud_Risk_Analytics.ipynb (Colab notebook)
- fraud_cleaned.csv (Power BI ready) and 7 summary CSV files
- Credit_Card_Fraud_Risk_Dashboard.pbix (Power BI dashboard)
- This project report and the GitHub README